① Să se determine $f: \mathbb{C} \to \mathbb{C}$ olomorfă, $f = P + iQ$ a.î. $Q(x,y) = f(ax+by)$, $f \in C^2$; $a, b \in \mathbb{R}^*$

$$\Delta Q = 0 \to \frac{\partial^2 Q}{\partial x^2} + \frac{\partial^2 Q}{\partial y^2} = 0$$

$$\frac{\partial Q}{\partial x} = f'(ax+by)\cdot a \qquad \frac{\partial^2 Q}{\partial x^2} = f''(ax+by)\cdot a^2$$

$$\frac{\partial Q}{\partial y} = f'(ax+by)\cdot b \qquad \frac{\partial^2 Q}{\partial y^2} = f''(ax+by)\cdot b^2$$

$$\Delta Q = 0 \Rightarrow a^2 f''(ax+by) + b^2 f''(ax+by) = 0$$

$$\Rightarrow (a^2+b^2) f''(ax+by) = 0 \Rightarrow f''(ax+by) = 0$$

$$\Rightarrow f'(ax+by) = k,\ k \in \mathbb{R}$$

Notăm $ax+by = \alpha \to f(\alpha) = k\alpha + d$ $k, d \in \mathbb{R}$

$$C\text{-}R \Rightarrow \begin{cases} \frac{\partial P}{\partial x} = \frac{\partial Q}{\partial y} = kb \xrightarrow[\text{prim } x]{\text{int.}} P(x,y) = kbx + c(y) \\ \frac{\partial P}{\partial y} = -\frac{\partial Q}{\partial x} = -ka \end{cases} \not\Longrightarrow$$

$$\Rightarrow \frac{\partial P}{\partial y} = c'(y) = -ka \Rightarrow c(y) = -kay + e \qquad k, e \in \mathbb{R}$$

$$P(x,y) = kbx - kay + e \ ; k, e \in \mathbb{R}$$

$$f(x+iy) = kbx - kay + e + i(kax + kby + d) \quad k, e, d \in \mathbb{R}$$

$$f(x+iy) = kb(x+iy) - ka(y - ix) + e + id$$

$f(x+iy) = kb \cdot z + i ka (x+iy) + e + id$

$f(x+iy) = kb \cdot z + i ka \cdot z + e + id = kz(b+ia) + e + id$

$k, e, d \in \mathbb{R}$

② Aflați

a) $Lu(\sqrt{3}-i) = ?$

$Lu(\sqrt{3}-i) = \ln 2 + i(\arg z + 2k\pi) \quad k \in \mathbb{Z}$

$\frac{z}{|z|} = \cos\theta + i \sin\theta$

$\frac{\sqrt{3}-i}{2} = \frac{\sqrt{3}}{2} - i \cdot \frac{1}{2}$ $\Rightarrow \theta \in C_{IV}$

$\theta = 2\pi - \frac{\pi}{6} = \frac{11\pi}{6}$

$Lu(\sqrt{3}-i) = \ln 2 + i\left(\frac{11\pi}{6} + 2k\pi\right), \quad k \in \mathbb{Z}$

b) $Lu\left(\frac{-1+i\sqrt{3}}{1+i}\right)$

$|-1+i\sqrt{3}| = \sqrt{1+3} = 2$

$\frac{-1+i\sqrt{3}}{2} = \cos\theta + i \sin\theta = \frac{-1}{2} + i \frac{\sqrt{3}}{2}$

$\theta = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$

$|1+i| = \sqrt{2}$

$\frac{1+i}{\sqrt{2}} = \cos\theta + i \sin\theta = \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2}$

$\theta = \frac{\pi}{4}$

$z_1 = |z_1|(\cos\theta_1 + i \sin\theta_1)$
$z_2 = |z_2|(\cos\theta_2 + i \sin\theta_2)$ $\Rightarrow \frac{z_1}{z_2} = \left|\frac{z_1}{z_2}\right|[\cos(\theta_1-\theta_2) + i \sin(\theta_1-\theta_2)]$

$$\frac{1+i\sqrt{3}}{1+i} = \frac{2}{\sqrt{2}}\left[\cos\left(\frac{2\pi}{3}-\frac{\pi}{4}\right)+i\sin\left(\frac{2\pi}{3}-\frac{\pi}{4}\right)\right]$$

$$\frac{1+i\sqrt{3}}{1+i} = \sqrt{2}\left[\cos\left(\frac{5\pi}{12}\right)+i\sin\left(\frac{5\pi}{12}\right)\right]$$

$$Ln\left(\frac{-1+i\sqrt{3}}{1+i}\right) = \ln\sqrt{2}+i\left(\frac{5\pi}{12}+2k\pi\right),\ k\in\mathbb{Z}$$

c) $\cos z = 2i$

$$\cos z = \frac{e^{iz}+e^{-iz}}{2} = 2i$$

$$e^{iz}+e^{-iz} = 4i \mid \cdot e^{iz}$$

$$e^{2iz}+1 = 4ie^{iz}$$

Notăm $e^{iz}=t$ $\Rightarrow t^2-4it+1=0$

$$\Delta = -16-4 = -20$$

$$t_{1,2} = \frac{4i \pm i\sqrt{20}}{2} = 2i \pm i\sqrt{5}$$

$$e^{iz} = 2i \pm i\sqrt{5}$$

$$\Rightarrow iz = \ln(2i \pm i\sqrt{5}) = \ln\left|(2\pm\sqrt{5})i\right| + i\left(\arg(2i\pm i\sqrt{5})+2k\pi\right),\ k\in\mathbb{Z}$$

$$\frac{(2+\sqrt{5})i}{2+\sqrt{5}} = i = 0+1\cdot i \Rightarrow \sin(2+\sqrt{5})i \Rightarrow \theta = \frac{\pi}{2}$$

$$\frac{(2-\sqrt{5})i}{\sqrt{5}-2} = -i = 0-1\cdot i \Rightarrow \sin(2-\sqrt{5})i \Rightarrow$$

$$\Rightarrow \theta = 2\pi - \frac{\pi}{2} = \frac{3\pi}{2}$$

$$iz = \ln(\sqrt{5}\pm 2) + i\left(\pm\frac{\pi}{2}+2k\pi\right)\ k\in\mathbb{Z}$$

③ Pentru ce valori $z \in \mathbb{C}$, $i^z \in \mathbb{R}$?

$$i^z = e^{z \ln i}$$

$$e^{z \ln i} = e^{(a+bi)\cdot i\left(\frac{\pi}{2}+2k\pi\right)} = e^{ai\left(\frac{\pi}{2}+2k\pi\right) - b\left(\frac{\pi}{2}+2k\pi\right)} \quad k \in \mathbb{Z}$$

$$e^{z \ln i} = e^{-b\left(\frac{\pi}{2}+2k\pi\right)} \cdot \left[\cos a\left(\pi/2+2k\pi\right) + i \sin a\left(\pi/2+2k\pi\right)\right] \in \mathbb{R}$$

$$\Leftrightarrow \sin a\left(\pi/2+2k\pi\right) = 0 \Rightarrow a\left(\pi/2+2k\pi\right) = q\pi, \; q \in \mathbb{Z}$$

$$\Rightarrow a = \frac{q}{\frac{1}{2}+2k}, \; q, k \in \mathbb{Z}$$

$$z = \frac{q}{\frac{1}{2}+2k} + ib$$

④ a) $\operatorname{ch} z - \operatorname{sh} z = 1+i$

$$\frac{e^z + e^{-z}}{2} - \frac{e^z - e^{-z}}{2} = 1+i$$

$$e^{-z} = 1+i \Rightarrow -z = \ln(1+i)$$

$$-z = \ln\sqrt{2} + i\left(\arg z + 2k\pi\right), \; k \in \mathbb{Z}$$

$$-z = \ln\sqrt{2} + i\left(\frac{\pi}{4} + 2k\pi\right), \; k \in \mathbb{Z}$$

b) $(1+i\sqrt{3})^i$

$$(1+i\sqrt{3})^i = e^{i \ln(1+i\sqrt{3})}$$

$$= \left\{ e^{i\left[\ln|1+i\sqrt{3}| + i\left(\arg(1+i\sqrt{3}) + 2k\pi\right.\right.}, \; k \in \mathbb{Z} \right\}$$

$$|1+i\sqrt{3}| = \sqrt{1+3} = \sqrt{4} = 2 \qquad \frac{1}{2} + i\frac{\sqrt{3}}{2} = \cos\theta + i\sin\theta$$

$$(1+i\sqrt{3})^i = \left\{ e^{i\left[\ln 2 + i\left(\frac{\pi}{3}+2k\pi\right)\right]},\ k\in\mathbb{Z} \right\}$$

$$= \left\{ e^{i\ln 2} \cdot e^{-\left(\frac{\pi}{3}+2k\pi\right)},\ k\in\mathbb{Z} \right\}$$

$$= \left\{ \left[\cos(\ln 2) + i\sin(\ln 2)\right] \cdot e^{-\left(\frac{\pi}{3}+2k\pi\right)},\ k\in\mathbb{Z} \right\}$$

c) $e^{\sqrt{i}}$

$$\sqrt{i} = i^{1/2} = e^{\frac{1}{2}\ln i} = e^{1/2\, i\left(\frac{\pi}{2}+2k\pi\right)} = e^{i\left(\frac{\pi}{4}+k\pi\right)}$$

$$\sqrt{i} = \cos\left(\frac{\pi}{4}+k\pi\right) + i\sin\left(\frac{\pi}{4}+k\pi\right),\ k\in\{0,1\}$$

$$k=0 \longrightarrow \sqrt{i} = \cos\left(\frac{\pi}{4}\right) + i\sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2} + i\frac{\sqrt{2}}{2} = \frac{1+i}{\sqrt{2}}$$

$$k=1 \longrightarrow \sqrt{i} = \cos\left(\frac{5\pi}{4}\right) + i\sin\left(\frac{5\pi}{4}\right) = \frac{-1-i}{\sqrt{2}}$$

$$\sqrt{i} = \frac{\pm 1 \pm i}{\sqrt{2}}$$

$$e^{\sqrt{i}} = e^{\pm\frac{1+i}{\sqrt{2}}}$$

d) $\operatorname{th} z = 2$

$$\frac{\operatorname{sh} z}{\operatorname{ch} z} = \frac{e^z - e^{-z}}{2} \cdot \frac{2}{e^z + e^{-z}} = \frac{e^z - e^{-z}}{e^z + e^{-z}} = 2$$

$$e^z - e^{-z} = 2e^z + 2e^{-z} \Rightarrow e^z = -3e^{-z} \quad | \cdot e^z$$

$$e^{2z} = -3$$

$$\Rightarrow 2z = \left\{ \ln|-3| + i(\pi + 2k\pi),\ k\in\mathbb{Z} \right\} = \left\{ \ln 3 + i(\pi+2k\pi),\ k\in\mathbb{Z} \right\}$$

$$z = \frac{1}{2}\left\{ \ln 3 + i(\pi+2k\pi),\ k\in\mathbb{Z} \right\}$$

e) $\sin z + i\operatorname{sh} z = 0$

$\boxed{\sin(iz) = i\operatorname{sh} z}$

$$\sin z + i\operatorname{sh} z = \sin z + \sin(iz) = 2 \cdot \sin\frac{z+iz}{2} \cos\frac{z-iz}{2}$$

$$\text{I} \quad \sin\frac{z+iz}{2} = 0 \Rightarrow \frac{z+iz}{2} = k\pi \Rightarrow z(1+i) = 2k\pi$$

$$z = \frac{2k\pi}{1+i} \qquad k \in \mathbb{Z}$$

$$\text{II} \quad \cos\frac{z-iz}{2} = 0 \Rightarrow \frac{z-iz}{2} = k\pi + \frac{\pi}{2} \Rightarrow z(1-i) = 2k\pi + \pi$$

$$z = \frac{2k\pi + \pi}{1-i} \qquad k \in \mathbb{Z}$$

5) Calculați următoarele integrale

a) $$\int_{8+\pi i}^{8-3\pi i} e^{z/2}\, dz = 2e^{z/2}\Big|_{8+\pi i}^{8-3\pi i} = 2\left(e^{4-\frac{3}{2}\pi i} - e^{4+\frac{1}{2}\pi i}\right)$$

$$= 2\left[e^4\left(\cos\frac{3\pi}{2} - i\sin\frac{3\pi}{2}\right) - e^4\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)\right]$$

$$= 2 \cdot e^4\left(0 - (-i) - 0 - i\right) = 0$$

b) $$\int_{|z|=1} e^{-z^2}\, dz$$

! $e^{-z^2} = e^{-(x+iy)^2} = e^{-x^2-2ixy+y^2}$

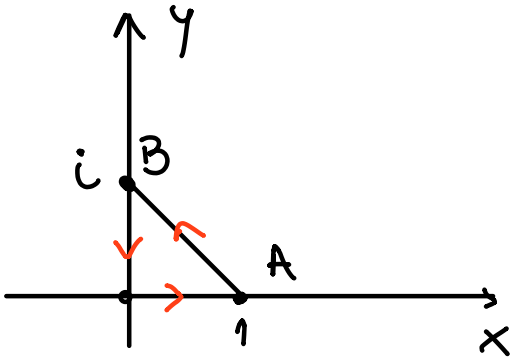
$= e^{-x^2+y^2} \cdot (\cos(2xy) - i\sin(2xy))$

$= \text{real}(P + iQ)$

e^{-z^2} funcție olomorfă, $|z|=1$ curbă închisă $\xrightarrow[\text{Cauchy}]{\text{Th.}}$ $\int = 0$

c) $\int_{\gamma} \operatorname{Im} z^2 \, dz$

γ este triunghiul cu vârfurile $0, 1, i$ parcurs în sens trigonometric



$z = x + iy \Rightarrow z^2 = x^2 + 2ixy - y^2$

$\operatorname{Im}(z^2) = \operatorname{Im}(x^2 + 2ixy - y^2) = 2xy$

Pe OA: $x = t$, $y = 0$ $\quad t \in [0,1] \xrightarrow{!} z(t) = t \longrightarrow \operatorname{Im}(z^2) = 0$

Pe AB: $x = 1 - t$, $y = t$ $\quad t \in [0;1]$ $\quad z(t) = 1 - t + it \quad t \in [0,1]$

Pe OB: $x = 0$, $y = t$ $\quad t \in [0,1] \xrightarrow{!} z(t) = it \longrightarrow \operatorname{Im}(z^2) = 0$

$z^2 = -t$

$$\int_{\gamma} \operatorname{Im} z^2 \, dz = \int_{OA} \operatorname{Im} z^2 \, dz + \int_{AB} \operatorname{Im} z^2 \, dz + \int_{BO} \operatorname{Im} z^2 \, dz$$

($\int_{OA} = 0$, $\int_{BO} = 0$)

$$\int_{\gamma} \operatorname{Im} z^2 \, dz = \int_0^1 \underbrace{2 \cdot (1-t) \cdot t}_{\operatorname{Im} z^2} \cdot \underbrace{(-1+i)}_{dz} dt = 2(-1+i) \int_0^1 t - t^2 \, dt$$

$$\int_{\gamma} \operatorname{Im} z^2 \, dz = 2(-1+i)\left(\frac{t^2}{2} - \frac{t^3}{3}\right)\Big|_0^1 = 2(-1+i)\left(\frac{1}{2} - \frac{1}{3}\right)$$

$$= 2(-1+i)\frac{3-2}{6_3} = (-1+i) \cdot \frac{1}{3}$$

d) $\int_{|z|=1} \frac{1}{3z-\bar{u}i}\,dz \overset{\text{Not. } \gamma}{=\!=}$

f olomorfă, dar nu $\bar{u}$ $\frac{\bar{u}i}{3}$ $\not\overset{\text{Th. Cauchy}}{\Longrightarrow}$ $\gamma=0$

$\frac{\bar{u}i}{3} \notin \text{Int}(|z|=1)$

e) $\int_{|z|=1} \sec\frac{z}{2}\,dz$

$\sec\frac{z}{2} = \frac{1}{\cos\frac{z}{2}}$

$\cos\frac{z}{2}=0 \Rightarrow \frac{e^{iz/2}+e^{-iz/2}}{2}=0 \Rightarrow e^{iz/2}+e^{-iz/2}=0$

$\Rightarrow e^{iz}+1=0 \Rightarrow e^{iz}=-1$

$iz = \{i(\bar{u}+2k\bar{u}),\ k\in\mathbb{Z}\} \Rightarrow z = \{\bar{u}+2k\bar{u},\ k\in\mathbb{Z}\} \not\Longrightarrow$

dar $|\bar{u}+2k\bar{u}|>1,\ \forall\, k\in\mathbb{Z}$

$\Rightarrow z=\{\bar{u}+2k\bar{u},\ k\in\mathbb{Z}\} \notin \text{Int}(|z|=1) \overset{\text{Th. Cauchy}}{\longrightarrow} \gamma=0$

f) $\int_{|z|=1} \frac{1}{\bar{z}}\,dz$ $\quad \gamma: |z|=1 \Rightarrow z(t)=e^{it} \quad t\in[0,2\bar{u})$

$$\gamma=\int_0^{2\bar{u}} \frac{1}{e^{-it}}\, i\cdot e^{it}\,dt = i\int_0^{2\bar{u}} e^{2it}\,dt = \frac{1}{2}e^{2it}\Big|_0^{2\bar{u}} = \frac{1}{2}\left(e^{4i\bar{u}}-e^0\right)$$

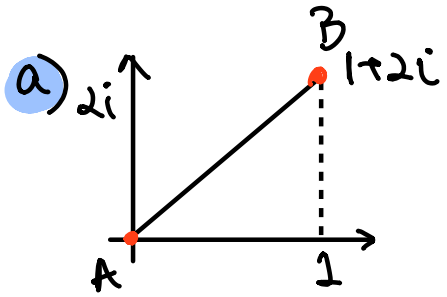
$$=\frac{1}{2}\left(\cos(4\bar{u})-i\sin(4\bar{u})-1\right)=\frac{1}{2}(1-0-1)=0$$

6) $\int_{\gamma} \operatorname{Re} z \, dz$, unde γ are traiectoria dată

Parametrizarea segmentului ce unește z_1 cu z_2

$$z(t) = (1-t)z_1 + tz_2 \quad t \in [0,1]$$

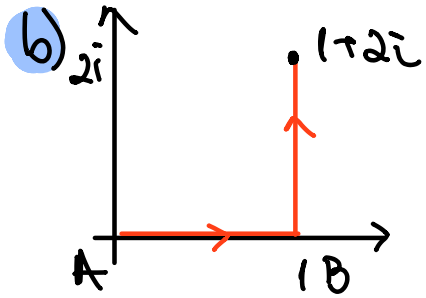
a)



$$AB: \quad z(t) = (1-t)(0+0\cdot i) + t\cdot(1+2i) \quad t \in [0,1]$$
$$z(t) = t(1+2i) = t + i\cdot 2t$$

$$\int_{\gamma} \operatorname{Re} z \, dz = \int_0^1 t\cdot(1+2i)\, dt = (1+2i)\frac{t^2}{2}\Big|_0^1 = (1+2i)\cdot\frac{1}{2}$$

b)



$$AB: \quad z(t) = (1-t)(0+0\cdot i) + t\cdot(1+0\cdot i)$$
$$z(t) = t \qquad t \in [0,1]$$

$$BC: \quad z(t) = (1-t)(1+0\cdot i) + t(1+2i) \quad t \in [0,1]$$
$$z(t) = (1-t) + t(1+2i) \quad t \in [0,1]$$
$$z(t) = 1 + 2it \qquad t \in [0,1]$$

$$\int_{\mathcal{L}} \operatorname{Re} z \, dz = \int_{AB} \operatorname{Re} z \, dz + \int_{BC} \operatorname{Re} z \, dz$$

$$\int_{\mathcal{L}} \operatorname{Re} z \, dz = \int_0^1 t \cdot 1 \, dt + \int_0^1 1 \cdot 2i \, dt = \frac{t^2}{2}\Big|_0^1 + 2i \cdot t\Big|_0^1 = \frac{1}{2} + 2i$$